

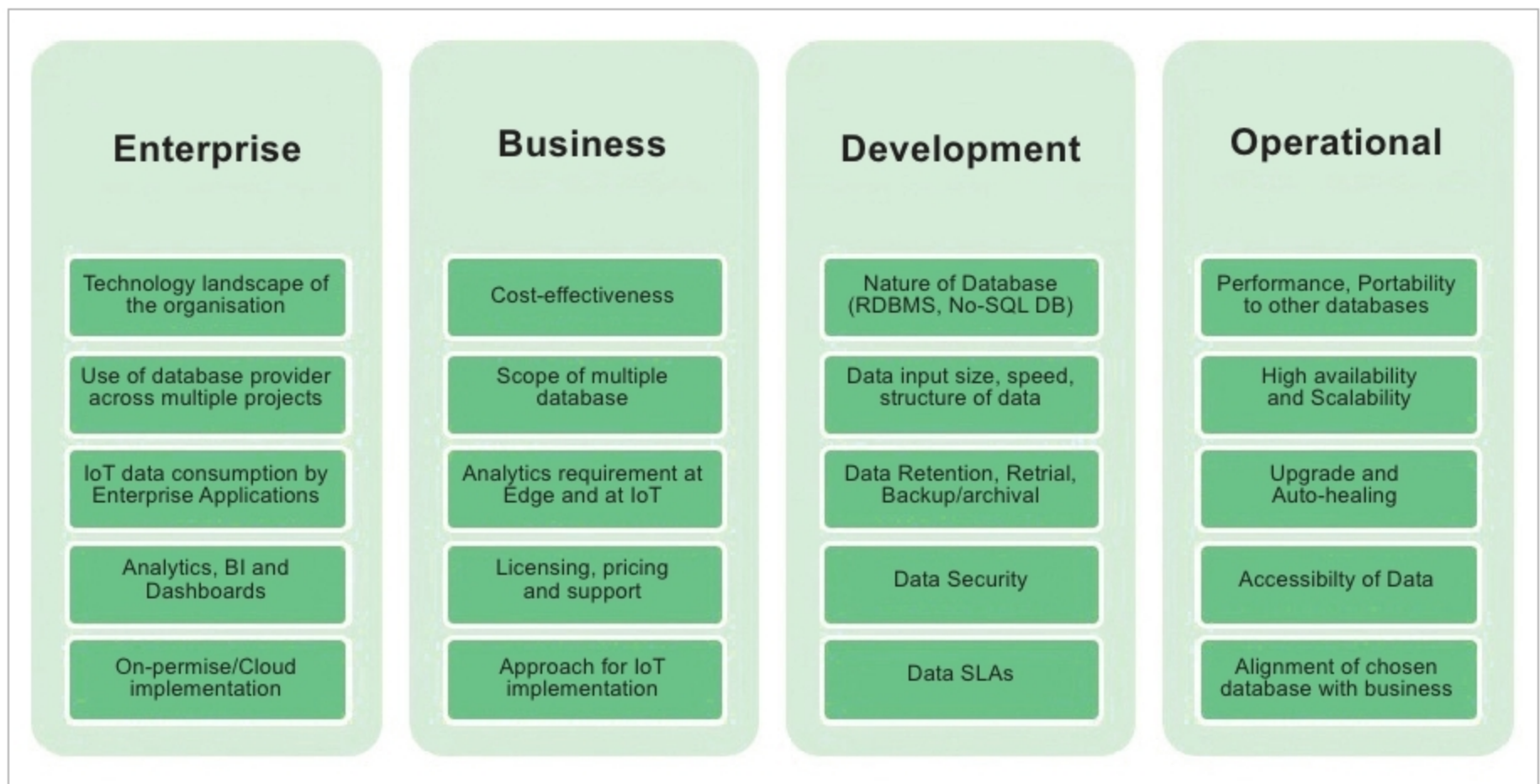


Figure 2: Factors to be considered when selecting a database

with minimum latency. The system analytics function collects the data from the edge server, and performs the data transformation and analytics operations. The database provides the commands to perform analytical computations on the data, and store the data for as long as required by the analytics engine.

IoT data streams convert the data to a common format and send it to enterprise systems. The database needs to perform the data transformation operations efficiently. The enterprise business intelligence (BI) function runs reports, queries and interfaces from historical data. The database needs to store data for a long period in a cost-effective manner. A warm database is a high-speed in-memory database that performs data read and write operations with the least latency. This database should be highly available and address disaster recovery requirements, providing real-time querying capabilities. A cold database is used to store the historical data of the IoT solutions. Typical databases could range from relational databases to a data lake.

Key open source databases for the IoT space

Every enterprise has a different business problem and IoT solutions can solve them in different ways, one of which is by growing a small setup to a large-scale implementation. The entire IoT implementation revolves around the idea of data collection/insertion through sensors and sending control back to devices. Selecting the right database for the various stages of IoT implementation in diverse use cases depends on multiple factors, as shown in Figure 2.

The following are some of the top open source databases available for IoT based applications.

MongoDB: This is a powerful and flexible open source database. It is a document-oriented and scalable NoSQL database. It supports features like indexes, range queries, sorting, aggregations and geospatial indexes. It also supports JSON to store and transmit information. MongoDB supports a rich query language for CRUD (create, read, update, delete) operations as well as data aggregation, text search and geospatial queries. As an example, Bosch has built its IoT suite on MongoDB.

InfluxDB: This is a time series database designed to handle high write and query loads. It provides an SQL-like query language called InfluxQL for interacting with data. InfluxDB has no external dependencies, and SQL-like queries are used for querying a data structure comprising measurements, series and points. Each point consists of varied key-value pairs called field set and timestamp. Values can be 64-bit integers, 64-bit floating points, strings and Booleans. Points are indexed by their time and tagset. InfluxDB stores data via HTTP, TCP and UDP. There is plugin support for other data ingestion protocols like Graphite, Collected, Open TSDB, etc.

BigchainDB: This is an embedded database for IoT devices. The assets created are stored as JSON documents in BigchainDB. It helps to convert physical objects into blockchain services by building blockchain-specific hardware that is compatible with any IoT device. It helps to define permissions for reading and writing rights into the IoT device. BigchainDB can be integrated with every IoT scenario where there is a requirement of immutability and tamper-proof storage of data assets, as well as